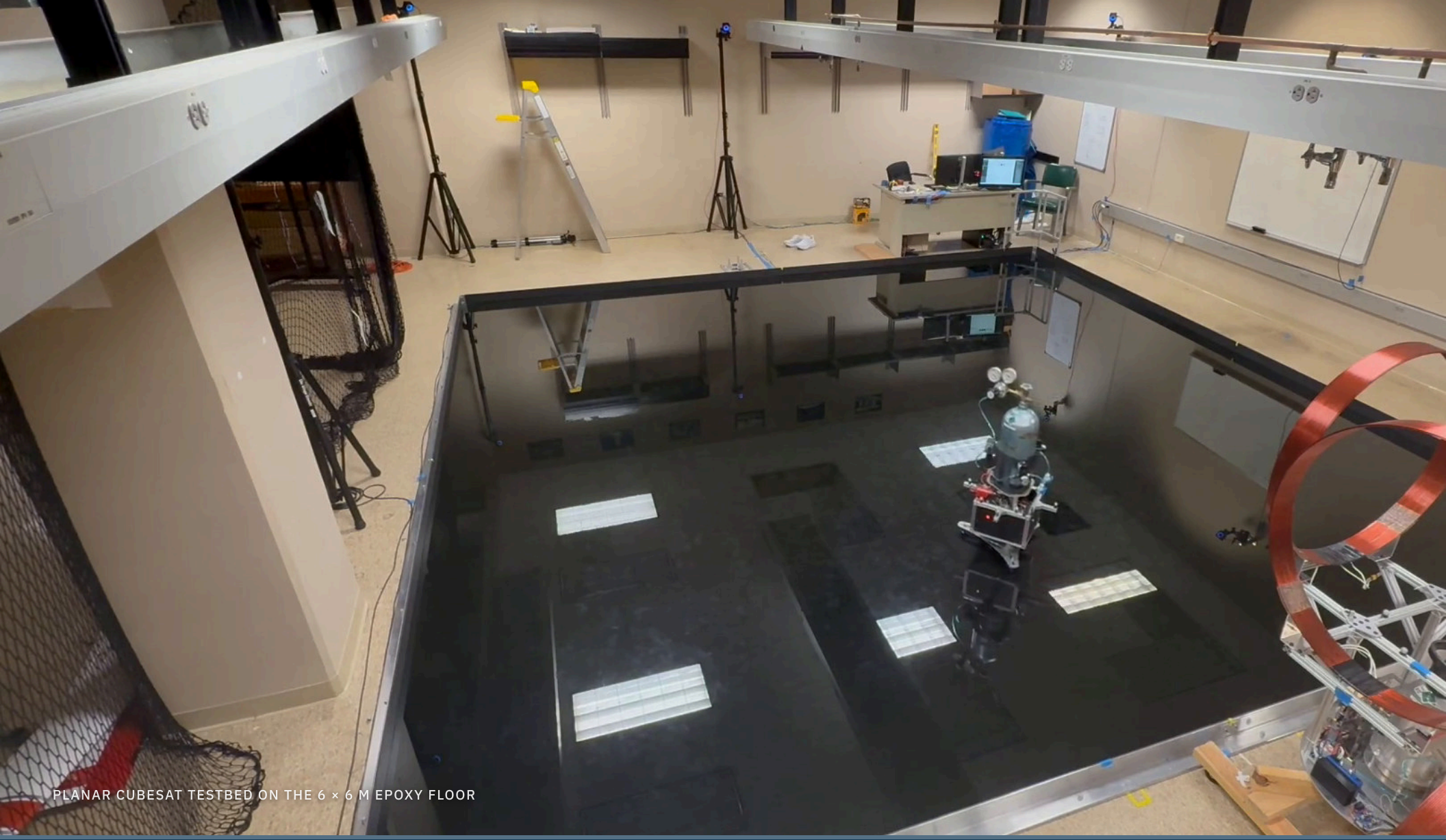


PLANAR CUBESAT TESTBED ON THE 6 × 6 M EPOXY FLOOR

MECHANICAL DESIGN PORTFOLIO

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M.S. THESIS · UNIVERSITY OF KENTUCKY, 2025 · NASA KY EPSCOR

Planar CubeSat Air-Bearing Testbed

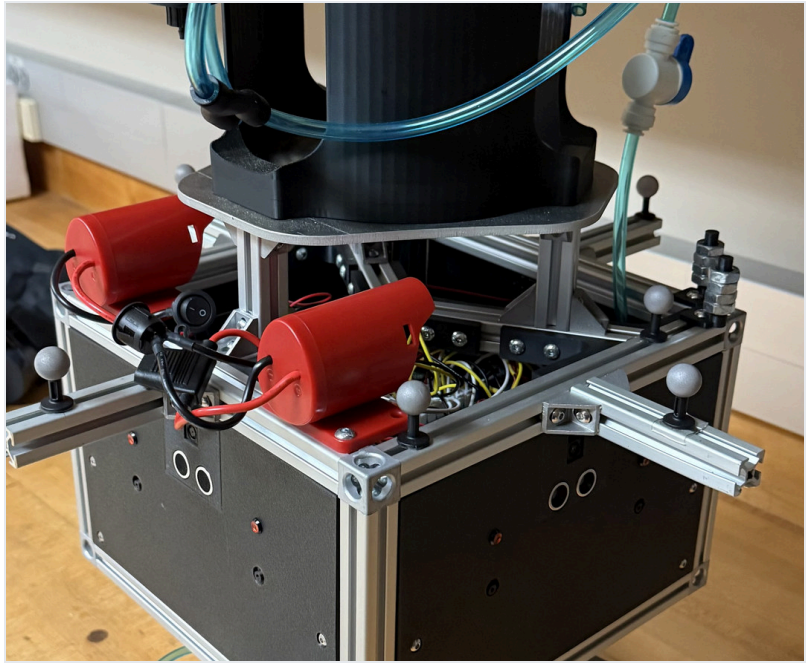
A ground testbed for simulation-to-hardware validation of mixed-integer MPC on a binary-thruster free flyer.

Satellites that inspect or assemble hardware in orbit need precise, fuel-aware planar maneuvering, but that behavior is expensive to test in space. I designed and built a ground testbed that floats a cubic platform on three air bearings over an epoxy floor, giving near-frictionless translation in x–y and rotation about the vertical axis. I used the testbed to validate a mixed-integer model-predictive controller for eight binary cold-gas thrusters against a digital twin.

I designed and built the entire testbed myself — the mechanical structure, pneumatics, electronics, and the control and simulation software. Advisors provided guidance.

The work was supported in part by NASA KY EPSCoR and is documented in my M.S. thesis, *Simulation-to-Hardware Validation of MPC for Binary Thruster 2D CubeSat Control*.

Thesis: https://uknowledge.uky.edu/me_etds/250/
Test video: *[Video link]*



Assembled platform with OptiTrack markers. The air cylinder is removed in this photograph; the 3D-printed tank mount, wiring, and triangular bearing plate are visible.

290 × 290 × 150 mm	3 × 40 mm	8 binary, 12 V	~2015 → 80 psi	~60 / ~40 psi	~23 kg · Fusion 360
BODY	POROUS-CARBON AIR BEARINGS	SOLENOID THRUSTERS	40 FT³ CYLINDER, THEN DUAL LEGS	BEARINGS / THRUSTERS	MASS · CAD

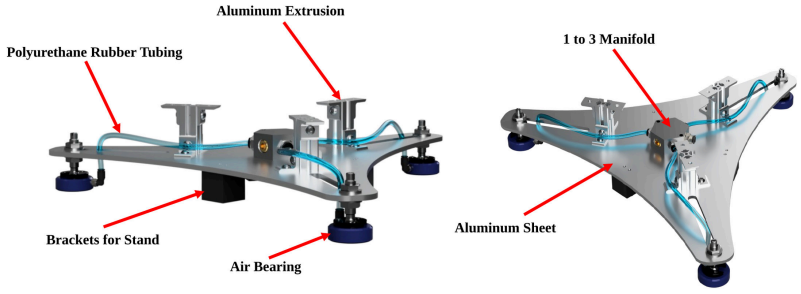
MECHANICAL DESIGN

Structure, packaging, and air-bearing support

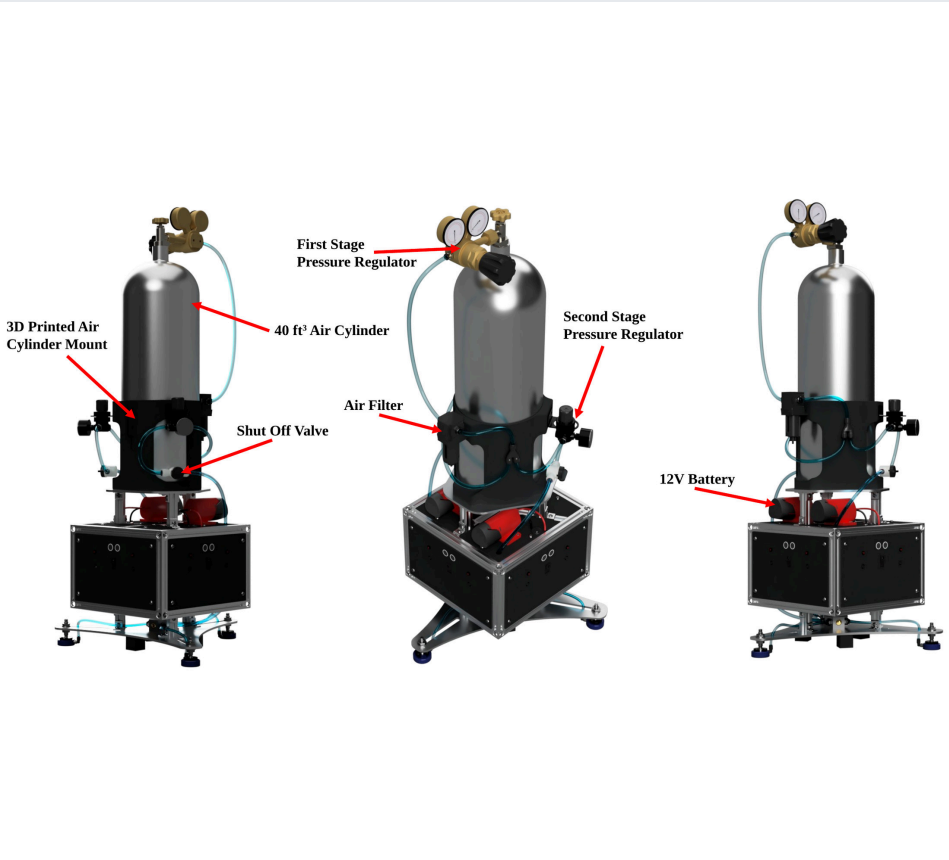
The chassis is a 290 × 290 × 150 mm frame of 2020 aluminum extrusion, assembled with corner brackets and T-nuts. Two internal decks and four side panels are 3 mm 3D-printed PLA, fastened with brass heat-set inserts. Level 1 carries the pneumatic manifold; Level 2, 50 mm above, carries the solenoids and electronics — isolating tubing from wiring and leaving a clear perimeter for the thrusters.

A triangular 6061 aluminum plate under the chassis holds three 40 mm porous-carbon air bearings. Radial grooves machined at 0°, 120°, and −120° let each bearing slide for load balance.

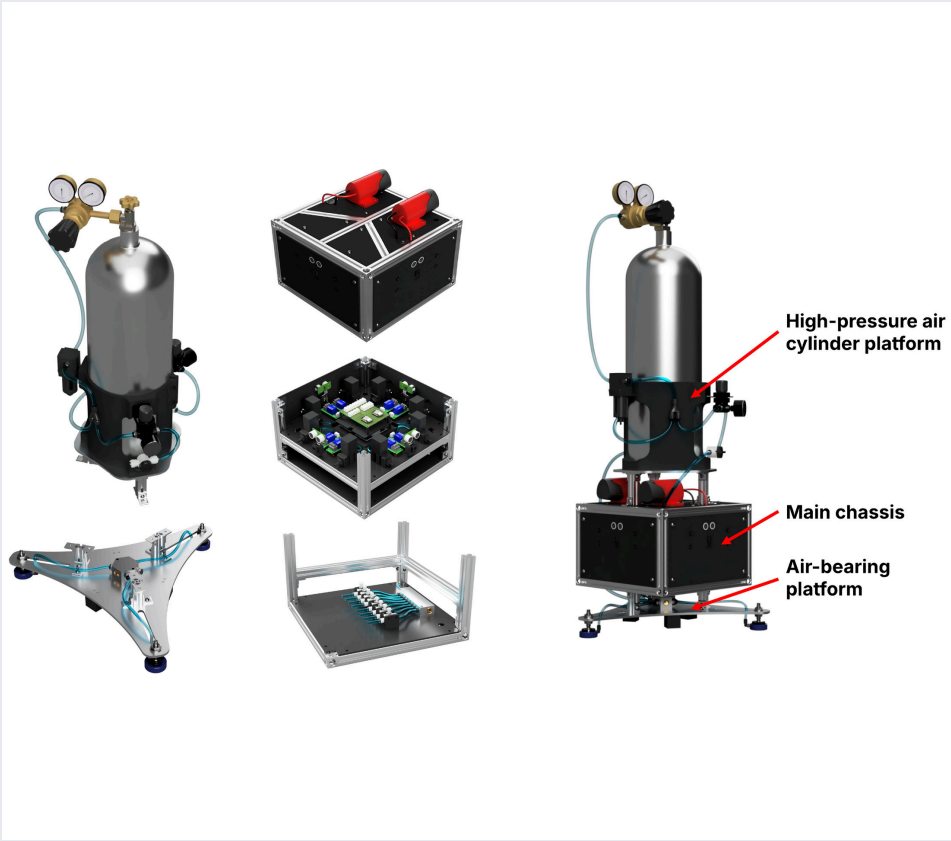
I built the design in Autodesk Fusion 360 and used the model for packaging, interference checks, hole locations and tolerances, and to export thruster geometry into the digital twin.



Air-bearing plate: 6061 aluminum sheet, three 40 mm bearings, 1-to-3 manifold, and machined radial adjustment grooves.



Complete assembly, three views. Cylinder, regulators, filter, shutoffs, M12 packs, chassis, and bearing plate.



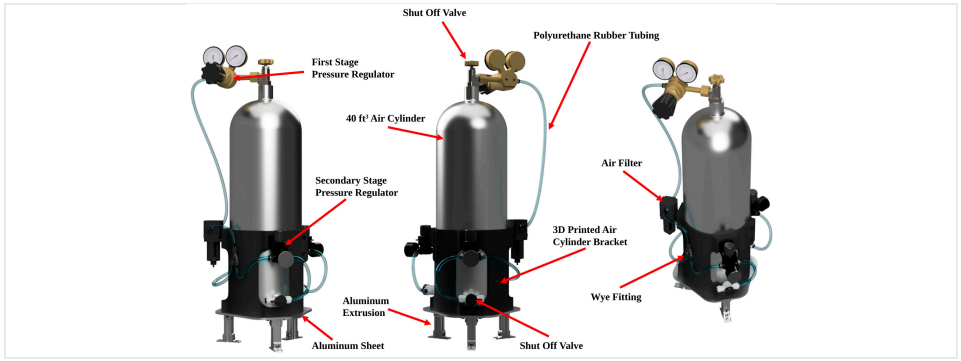
Subsystem montage: tank module, closed chassis, electronics deck, Level 1 pneumatics, air-bearing plate, and assembled vehicle.

CAD renders from the as-built Fusion 360 model

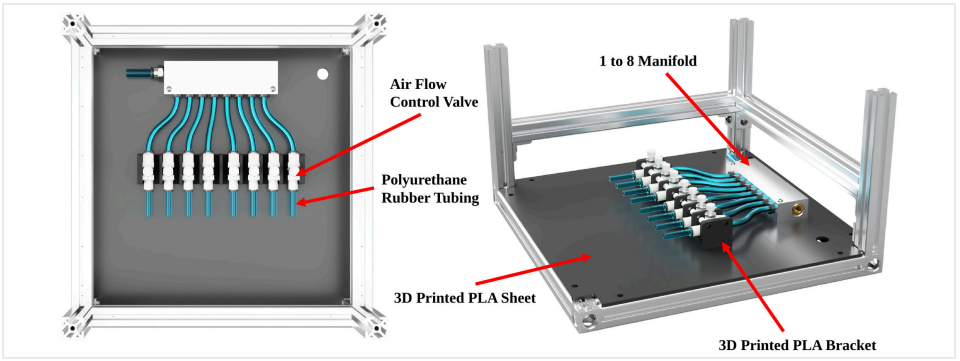
PNEUMATICS & THRUSTERS

Dual-pressure supply and in-situ thrust calibration

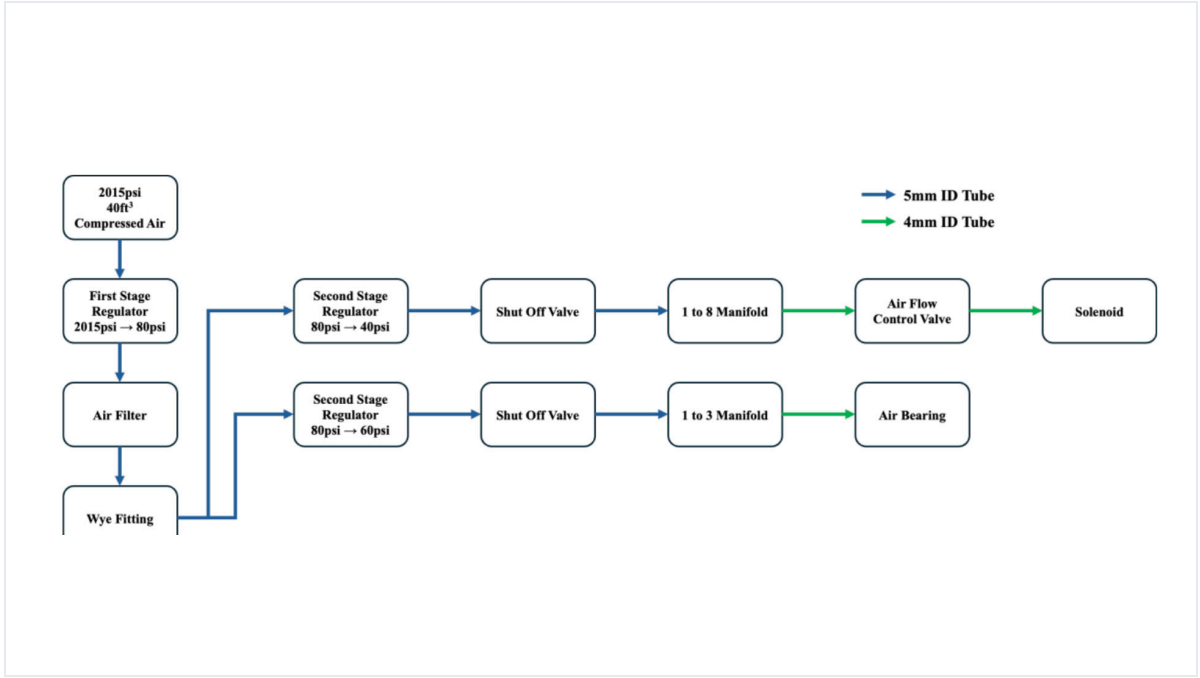
A 40 ft³ cylinder at about 2015 psi is regulated to about 80 psi, filtered, and split at a wye. A bearing leg is regulated to about 60 psi and a thruster leg to about 40 psi, with independent shutoffs so the two systems can be tested separately. Zero-grade air protects the porous-carbon bearing surfaces from moisture and oil. I designed a load-cell fixture and characterized each thruster in place on the vehicle, so line losses and regulator behavior were included. Steady-state thrust ranged from 0.428 to 0.484 N (mean 0.451 N), with up to 13.2% spread between thrusters. Those measured values are supplied to the controller rather than trimmed out in hardware.



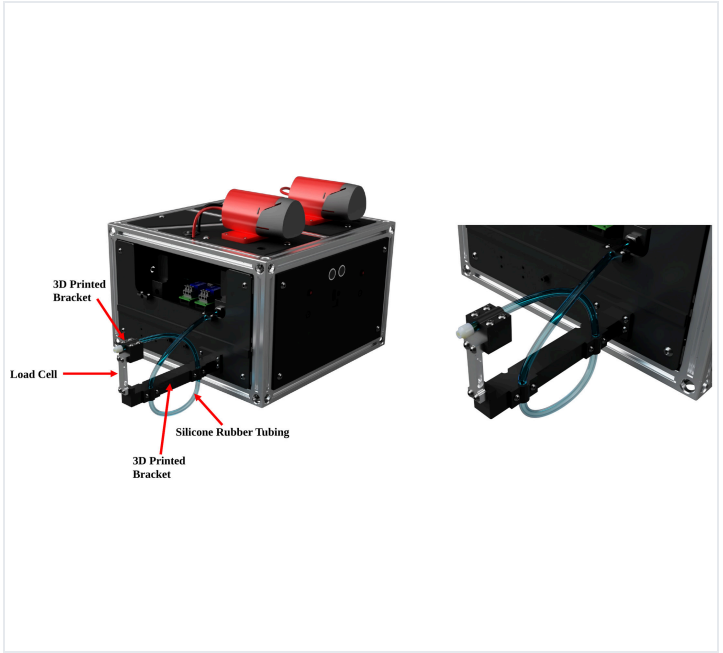
Pneumatic supply module: 40 ft³ cylinder, first- and second-stage regulators, filter, wye, and shutoffs on a 3D-printed bracket.



Level 1: 1-to-8 manifold, inline needle valves, and polyurethane tubing on a 3 mm 3D-printed PLA deck with heat-set inserts.



Pneumatic architecture (thesis Fig. 9): ~2015 psi cylinder → 80 psi → filter → wye → bearing leg (~60 psi) and thruster leg (~40 psi).



Load-cell fixture I designed for in-situ calibration. Thrust 0.428–0.484 N, mean 0.451 N; up to 13.2% spread is fed into the controller.

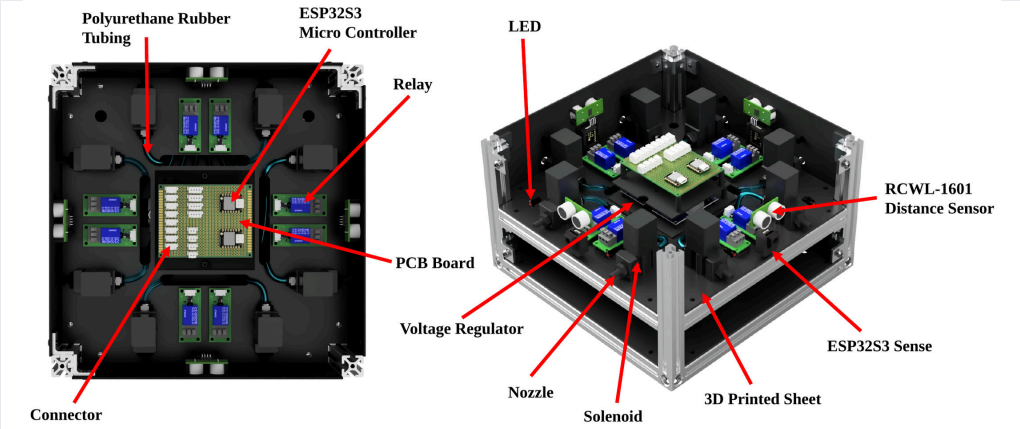
Flowchart from the thesis; CAD from the as-built model

ELECTRONICS & INTEGRATION

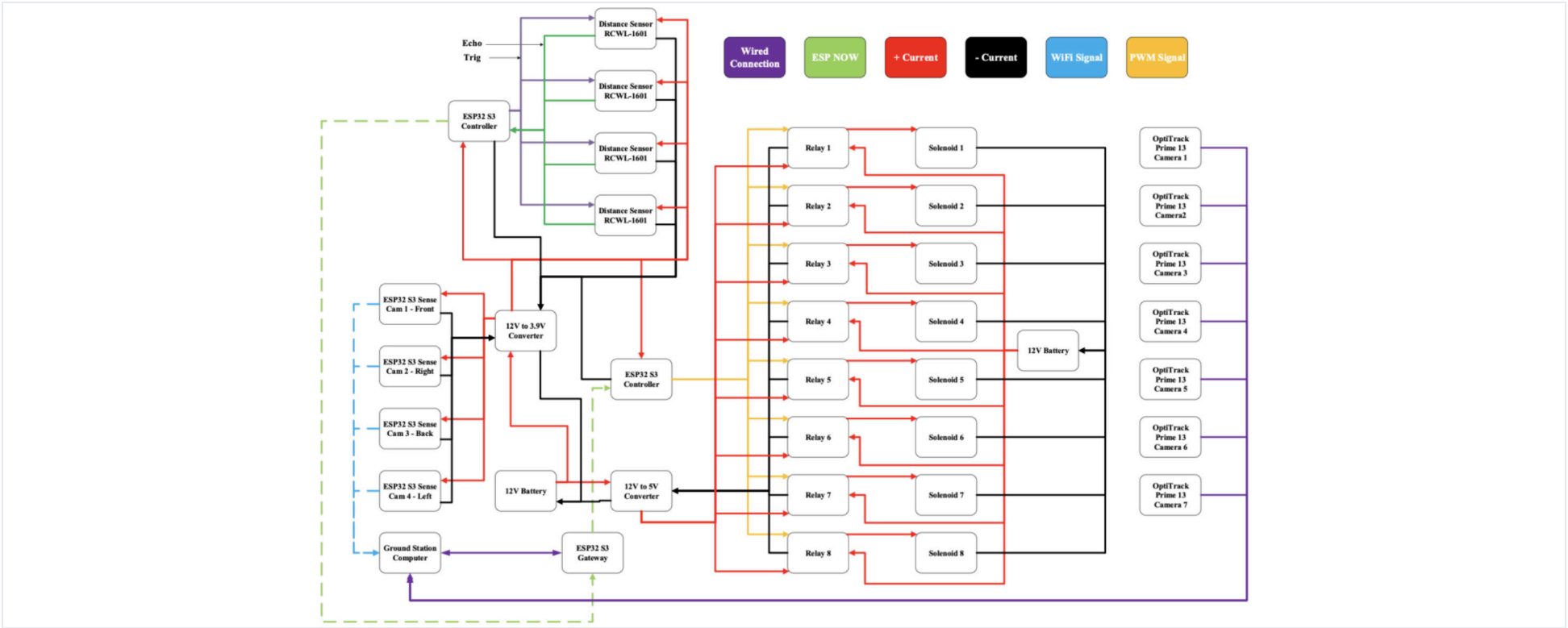
Onboard power, command path, and motion capture

The electronics are hand-built: a point-to-point soldered and wired prototype board serves as the power and signal hub. Two Milwaukee M12 packs separate the actuation bus (12 V to the solenoids and indicator LEDs) from the electronics bus (5 V and 3.9 V rails), so valve current spikes do not disturb the controllers.

Two XIAO ESP32-S3 microcontrollers sit on the platform — one drives eight relays that fire the solenoids; the other monitors auxiliary sensors. An ESP-NOW link carries 8-bit thruster commands from a ground-station gateway. Pose comes from seven OptiTrack Prime 13 cameras at 240 Hz around a 6 × 6 m epoxy floor. Mixed-integer MPC is solved on the ground station at 16.7 Hz; only on/off commands go to the vehicle.



Level 2: prototype board (labeled PCB Board in the model), ESP32-S3, relays, buck converters, solenoids, and side-panel sensors.



Electrical and communication diagram: split 12 V buses, DC-DC converters, onboard ESP32-S3 nodes, eight relays and solenoids, ESP-NOW gateway, and seven OptiTrack cameras.

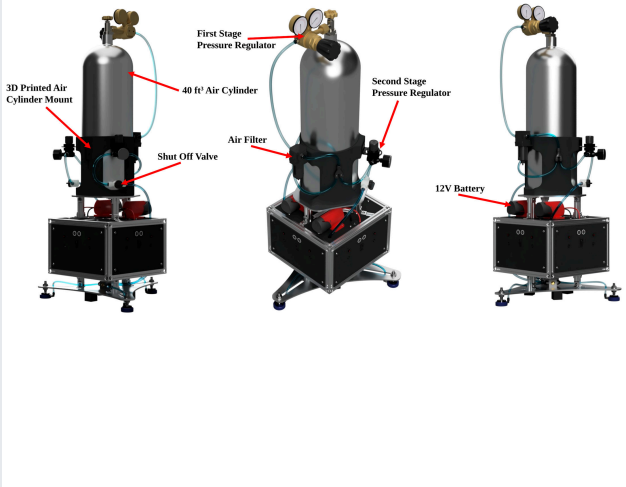
CAD from the as-built model; electrical diagram from the thesis (Fig. 12)

DESIGN ITERATIONS

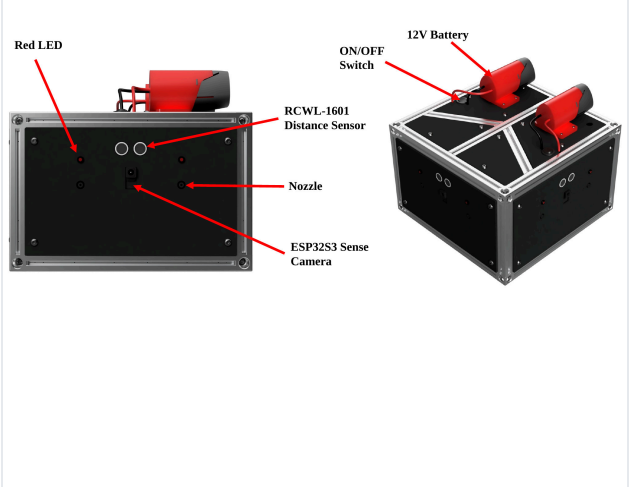
January 2025 concept versus the final build

A January 2025 design proposal captured an earlier concept. Before cutting hardware I also evaluated triangular, square, and hexagonal bearing layouts, alternative thruster placements, and different air-supply configurations in CAD. The square, eight-thruster layout was selected for redundancy without making the discrete control problem too large to solve in 60 ms. After assembly I adjusted the bearings along the machined grooves until support loads were within ±1.8% of equal, to reduce steady-state torque bias.

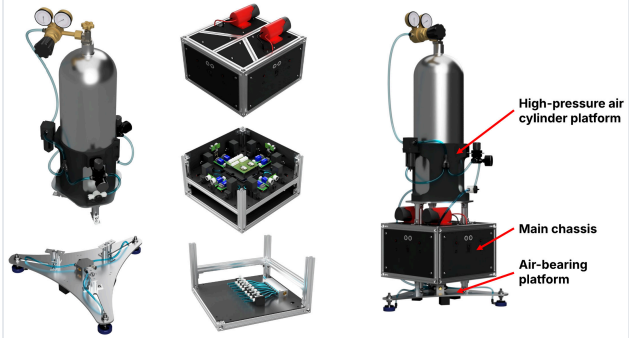
	January 2025 concept	Final build
Supply tank	20 ft³ at 2000 psi; first stage to 100 psi	40 ft³ at ~2015 psi; first stage to ~80 psi
Filter placement	After the wye, on the bearing line only	Before the split, protecting both legs
Power	Single 12 V battery	Two Milwaukee M12 packs (actuation and electronics buses)
Solenoid drivers	MOSFETs; PWM pulse width	Relays; binary on/off at 60 ms
Thrust matching	Needle valves used to equalize thrust	Valves left fixed; measured spread is compensated in the controller
Nozzles	3D-printed, customizable orifice	Commercial solenoids with an integrated ~2.5 mm orifice
Chassis	Yellow acrylic side panels; single internal bay	Black 3 mm 3D-printed PLA; two stacked decks



Final assembly. 40 ft³ supply, two M12 packs, black PLA panels.



Final closed chassis: side-panel LEDs, nozzles, ultrasonics, and cameras.



Final two-deck layout, with pneumatics isolated on Level 1.

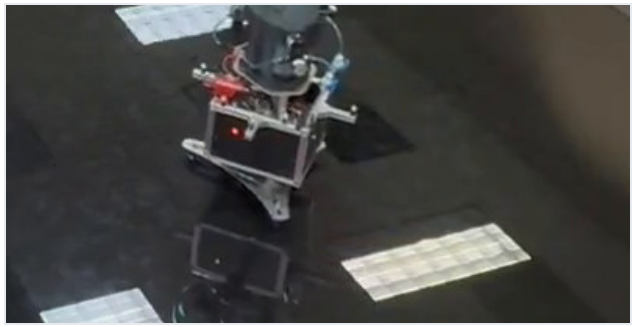
Comparison from the January 2025 proposal versus the as-built design. Images show the final model only.

TESTING & RESULTS

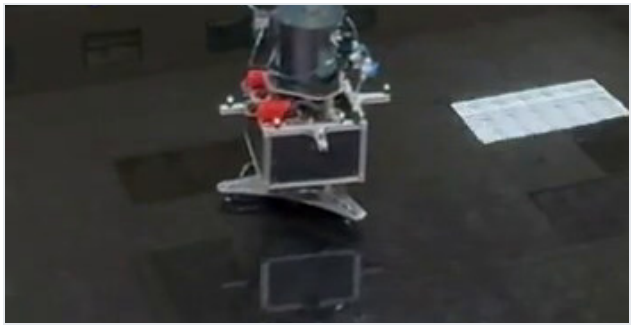
Hardware on the epoxy floor, compared with the digital twin

Closed-loop tests were run on the 6 × 6 m epoxy floor under OptiTrack. In the profile-following hardware test, mean tracking error was 0.179 m, with a final position error of 0.024 m and a final heading error of 0.51°. The mixed-integer MPC ran at 16.7 Hz (60 ms period). Mean solve time was 4.61 ms, and every cycle met the 60 ms deadline. Total thruster-on time agreed with the simulation to within 1.5%.

0.179 m	0.024 m	0.51°	16.7 Hz	4.61 ms	1.5%
MEAN TRACKING ERROR	FINAL POSITION ERROR	FINAL HEADING ERROR	MPC RATE (60 MS)	MEAN MPC SOLVE TIME	ON-TIME VS. SIMULATION



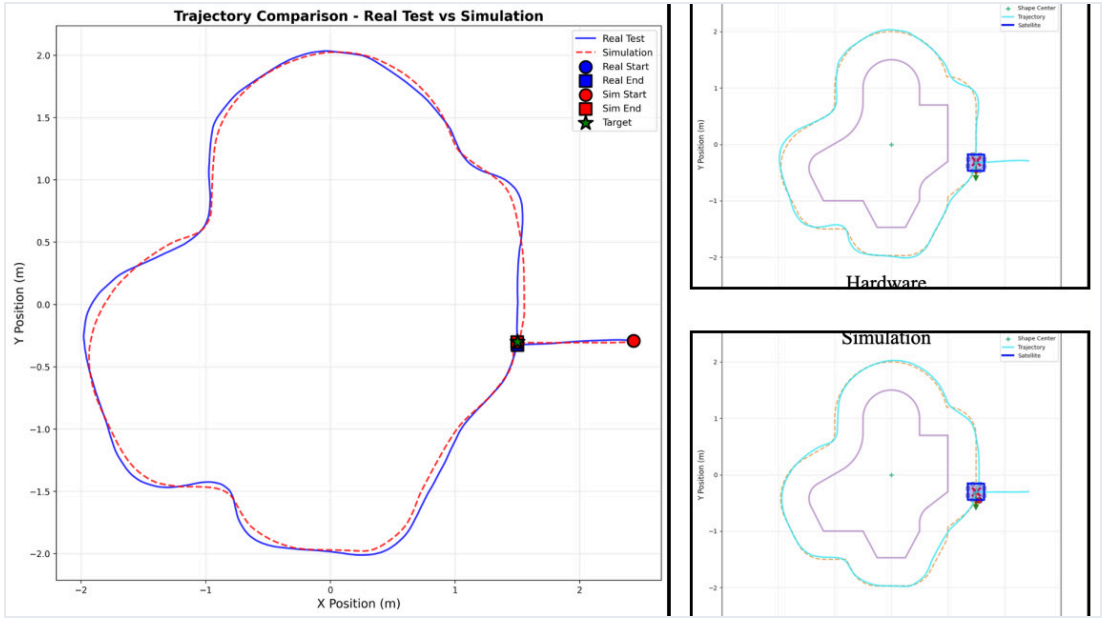
Overhead still, mid-run on the epoxy floor.



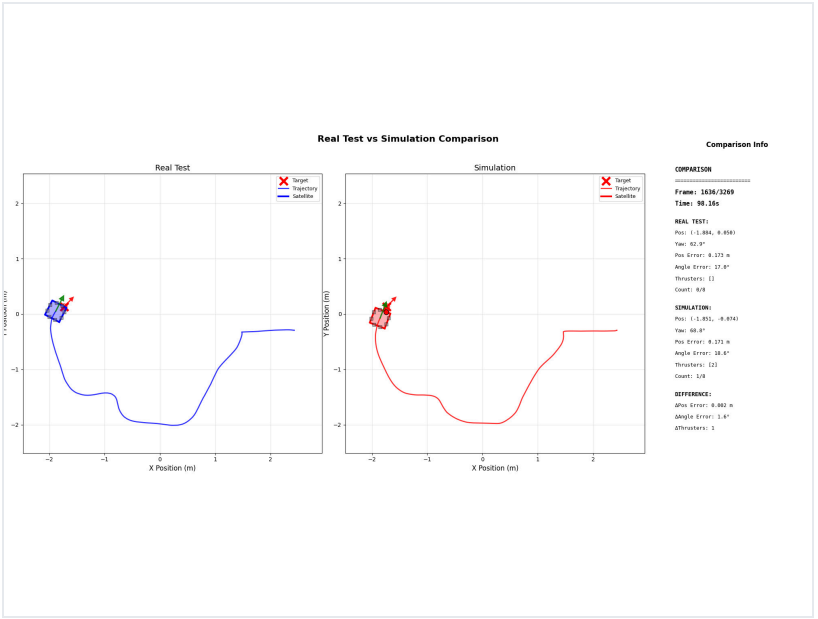
Same run, a few seconds later.



Later in the same test.



Trajectory comparison, hardware (blue) versus simulation (red dashed), profile-following mission.



Hardware vs. simulation comparison visualizer, mid-run overlay.

OTHER EXPERIENCE

Additional mechanical design work

The pages below are reserved for projects that will be added once material is ready. Layout, typography, and caption style will match the testbed section.

[Add Blue Lagoon project here]

Title, role, short description, and photos or CAD will go here.

[Add another project here]

A second project, set in the same layout as the card on the left.